VECTO 4.x

Vehicle Energy Consumption Calculation Tool

Release Notes

VECTO 0.11.2.3456 Development Version

18.06.2024

Vehicle Energy Consumption Calculation Tool

Release Notes

VECTO 3rd Amendment Development Version 18.06.2024

- Features
  - CodeEU-196: Implement wheel bearings

    Input for Engineering mode is available via the GUI & JSON file. In the vehicle file (.vveh) in an entry of the "Axles" array, you can add the field "WheelEndFriction" and assign it a numerical value.
    Input for Declaration mode is available via a temporary XSD (VectoDeclarationDefinitions.DEV.2.6.xsd). Sample file in folder: Generic Vehicles\Declaration Mode\Wheel_Bearings

  - CodeEU-643: added support for fuel cell IEPC

    Added new simulation job type for IEPC vehicles with fuel cells. This feature is only supported in engineering mode. The GUI has been amended to allow the creation/modification of this new job type. A sample job has been added to the folder 'Generic Vehicles\Engineering Mode\GenericIEPC - FCHV'.

VECTO 3rd Amendment Development Version 18.06.2024

- Bugfixes
  - CodeEU-236: Fuel Cell Simulation Aborts when Charging Power of Battery is too low
  - CodeEU-209: VECTO-0.11.1.3228-DEV crash in declaration mode
  - CodeEU-207: Writing XML Reports fails for vehicles without battery

VECTO 0.11.1.3382 Development Version

03.11.2023

Vehicle Energy Consumption Calculation Tool

Release Notes

VECTO 3rd Amendment Development Version 03.11.2023

- Changes
  - Enabled the usage of multiple Fuel Cell Systems (FCS)
    - Up to two FCS Strings
    - Up to three FCS in each FCS String
  - New naming convention:
    - FCS: Single Fuel Cell System, described with MassFlowMap (.vfcm) and power limits container in the Fuel Cell System File (.vfcc).
    - FCS String: Defined in the .vveh file, can contain up to three identical FCS
    - CFCS: Composite Fuel Cell System, a virtual fuel cell system containing up to two FCS String.
  - Changes in input files due to new naming convention
  - Optimized power distribution among FCS Strings and between individual FCS inside a string.(Explained in more detail in the user manual)
  - Allow operating a FCS below its minimum power via time splitting
  - Removed On/Off – Hysteresis
  - Removed Gradient Power Change

VECTO 0.11.0.3193 Development Version

29.09.2023

Vehicle Energy Consumption Calculation Tool

Release Notes

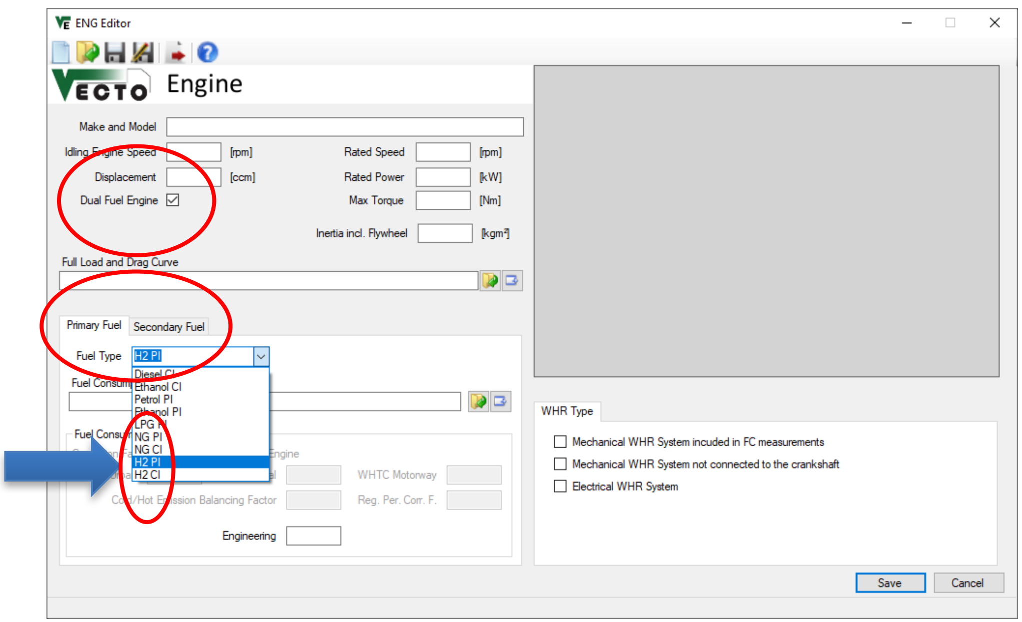
First VECTO beta release for the 3rd Amendment

H2 Internal Combustion Engines

TUG

H2 ICE

- New fuel types implemented „H2 PI“ and „H2 CI“ (i.e. H2-Diesel dual-fuel)
- Well-known ICE input data structure unchanged, no new elements



Fuel characteristics used

FuelType	FuelDensity [kg/m3]	CO2 per FuelWeight [kgCo2/kgFuel]	CO2 per FuelWeight VTP [kgCO2/kgFuel]	NCV_stdEngine [kJ/kg]	NCV_stdVecto [kJ/kg]
H2 PI	<empty>	0.00	0.00	120000	120000
H2 CI	<empty>	0.00	0.00	120000	120000

First VECTO beta release for the 3rd Amendment

Fuel Cell Hybrid Vehicles (FCHV)

TUG

Fuel cell hybrid vehicles (FCHV) – Op. strategy (1/5)

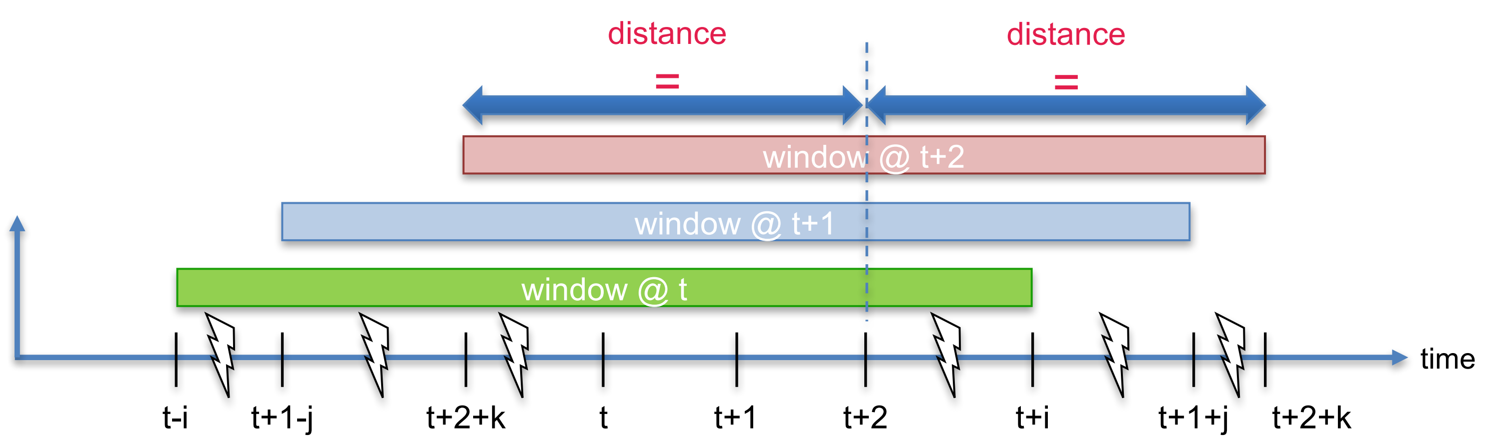
- FCHV generic operation strategy (Charge Sustaining Mode) implemented as presented in TF FCS #15 with following basic principle:
  - Maximum phlegmatisation of FCS by operation strategy
  - FCS needs to provide average electric power demand over longer horizon
  - Longest possible averaging horizon depends on battery size and specific vehicle-cycle combination
    → *The smaller the battery or the greater the electric energy demand of the vehicle, the smaller the averaging horizon will get leading to a more dynamic operation of the FCS.*

Fuel cell hybrid vehicles (FCHV) – Op. strategy (2/5)

- **Basic approach for implementation of FCS operation in "CS" Mode in 4 steps**
  1. Pre-simulation run as PEV to determine electric power demand for propulsion and auxiliaries
     a) Power limit of "special" battery defined as sum of respective maximum powers (BAT + FCS)
  2. Resulting total electric power from step 1 will be averaged over a certain **fixed** window size which is distance based (details next slide)
     a) Buffering of energy is limited by battery size, SOC limits min/max are not allowed to be exceeded
     b) Using largest possible window size without violating SOC limits in any window
     c) Adaption of start SOC to specific mission in order to maximize usage of battery as buffer
  3. Actual simulation run as FCHV in-the-loop where FCS is operated following a fixed power trace to cover average electric power demand determined from pre-processing in step 2 (battery covers instantaneous deviations from average electric power demand)
  4. ΔSOC correction in post-processing to account for small deviations from neutral SOC behaviour over cycle (e.g. variations in auxiliary power, small deviations in battery losses etc.)

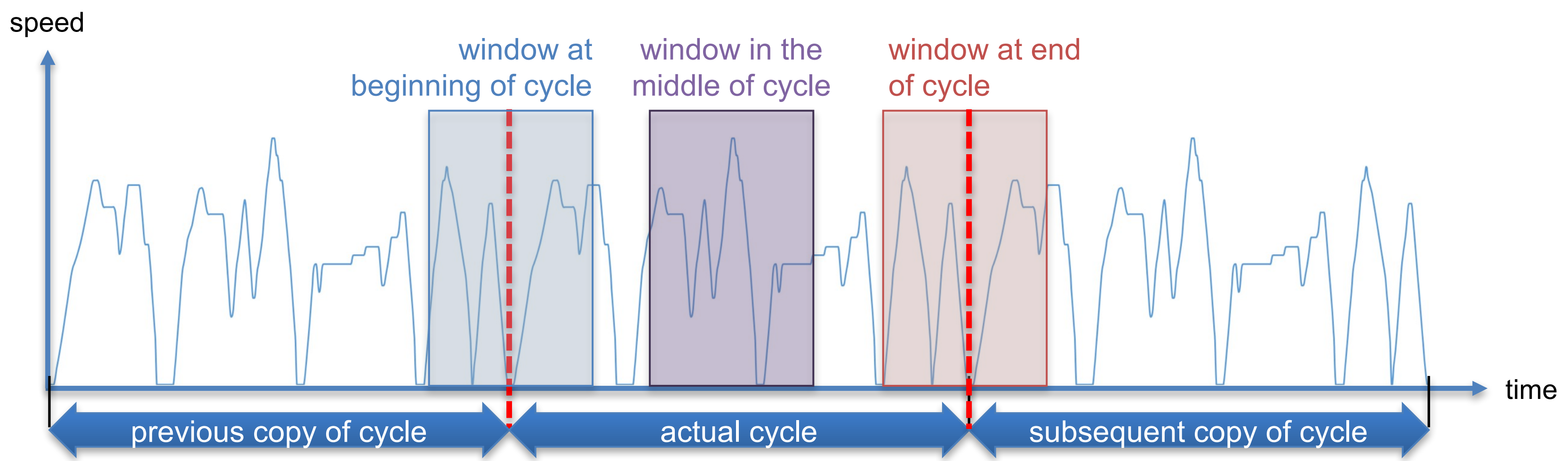
Fuel cell hybrid vehicles (FCHV) – Op. strategy (3/5)

- **Averaging and window size explained (1/2)**
  - The averaging window is placed distance-symmetrically over each actual time step from the pre-simulation
  - Window size is determined by total distance within window



Fuel cell hybrid vehicles (FCHV) – Op. strategy (4/5)

- **Averaging and window size explained (2/2)**
  - Maximum window size is given by complete cycle, thus cycle is concatenated 3 times in a row
  - For time steps near the beginning or end of the cycle, the windows expand over the actual cycle and continue at the adjacent parts at the end or beginning.

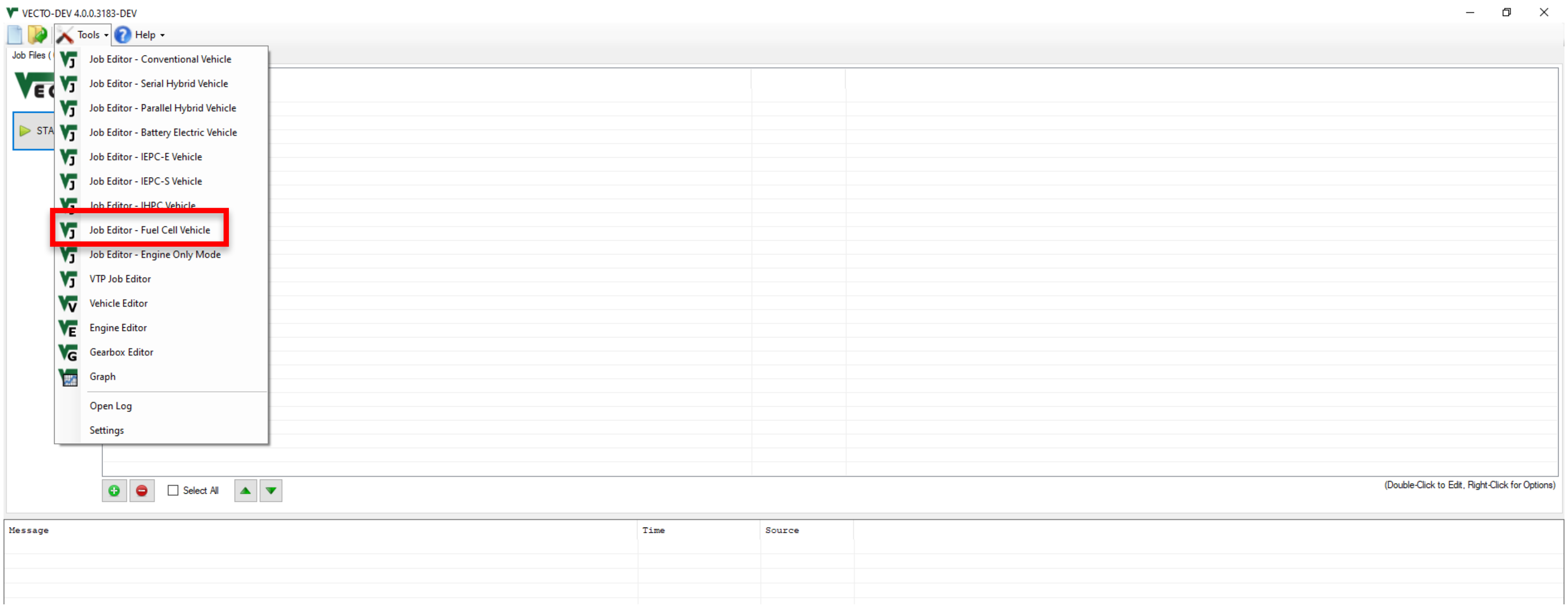


Fuel cell hybrid vehicles (FCHV) – Op. strategy (5/5)

- **Actual simulation run as FCHV and applicable boundary conditions**
  - In case average power demand over total cycle is higher than FCS maximum power:
    → defined error message and simulation is aborted
    (vehicle configuration not reasonable for this mission, should not occur in practice)
  - In case average power in a certain timestep is lower than FCS minimum power:
    → FCS is switched off
    → missing energy from FCS for those timesteps will be accounted for in post-processing via ΔSOC correction

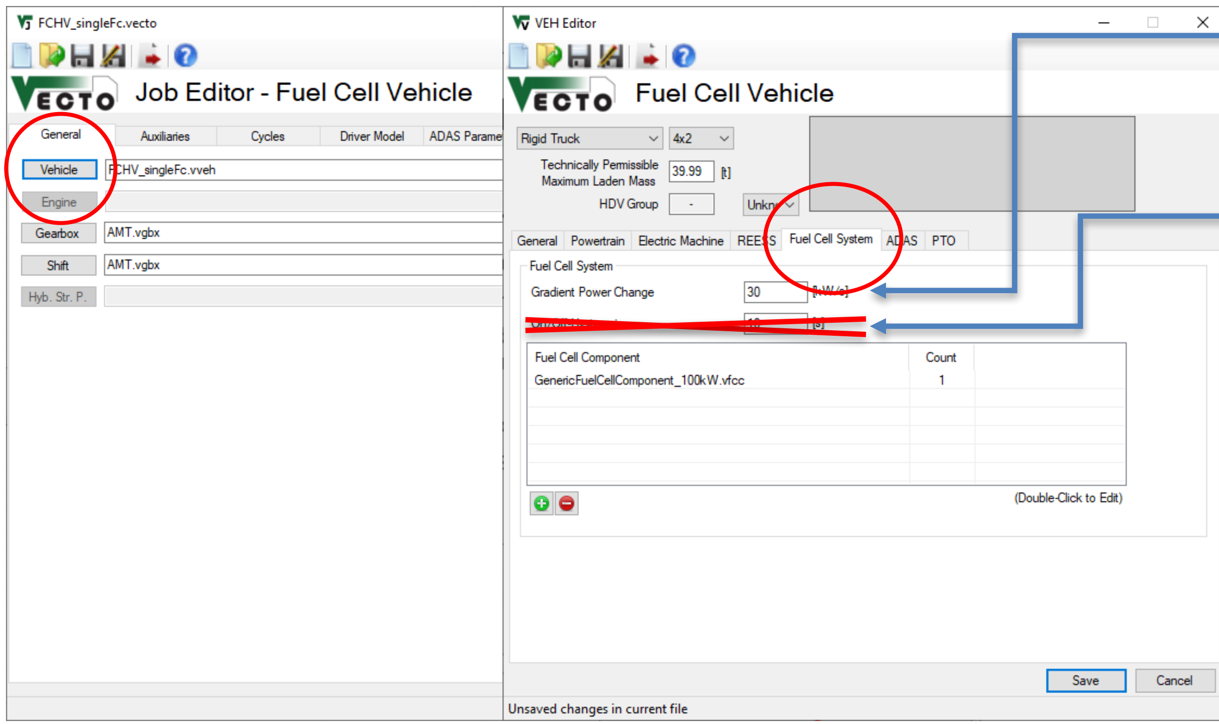
Fuel cell hybrid vehicles (FCHV) – VECTO features (1/5)

- New powertrain type of FCHV added to Job Editor



Fuel cell hybrid vehicles (FCHV) – VECTO features (2/5)

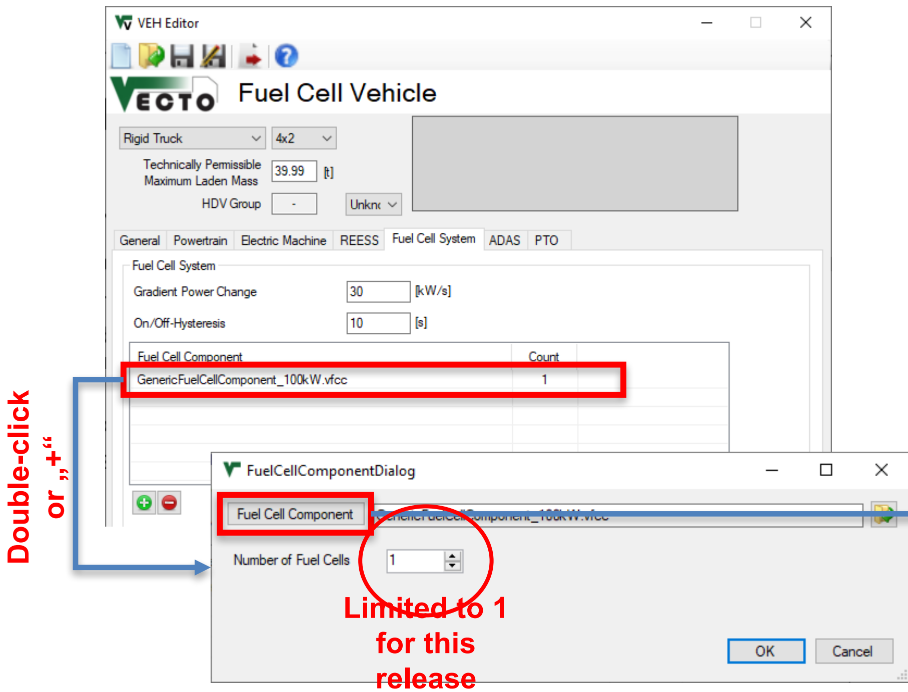
- FCS to be parameterized via specific „Fuel Cell System“ tab in „Vehicle“ editor

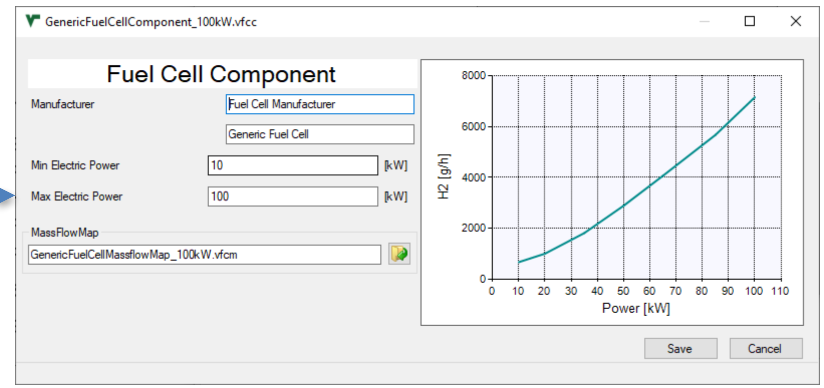


- Power gradient limits change in FCS power over time
- ~~On/off-Hysteresis is minimum time that FCS stays off or on after state change occurs~~
  → not required anymore after insights gained during work on implementation of multiple FCS

Fuel cell hybrid vehicles (FCHV) – VECTO features (3/5)

- FCS to be parameterized via specific „Fuel Cell System“ tab in „Vehicle“ editor

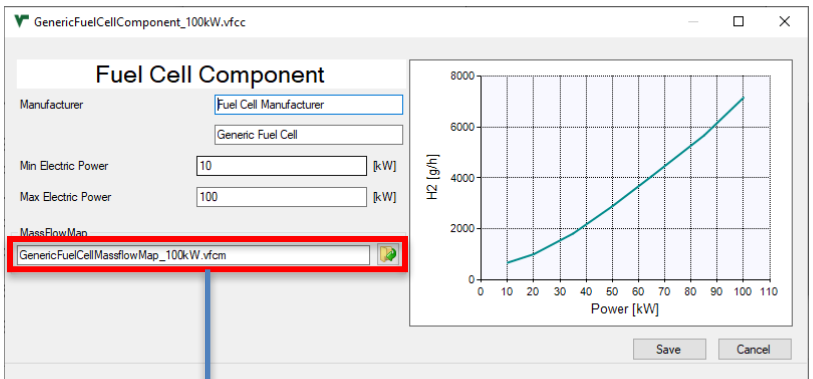




- Max / min power limits operation range of FCS component
- FCS mass flow map needs to cover the indicated range

Fuel cell hybrid vehicles (FCHV) – VECTO features (4/5)

- FCS to be parameterized via specific „Fuel Cell System“ tab in „Vehicle“ editor



Input data structure for FCS H2 mass flow map

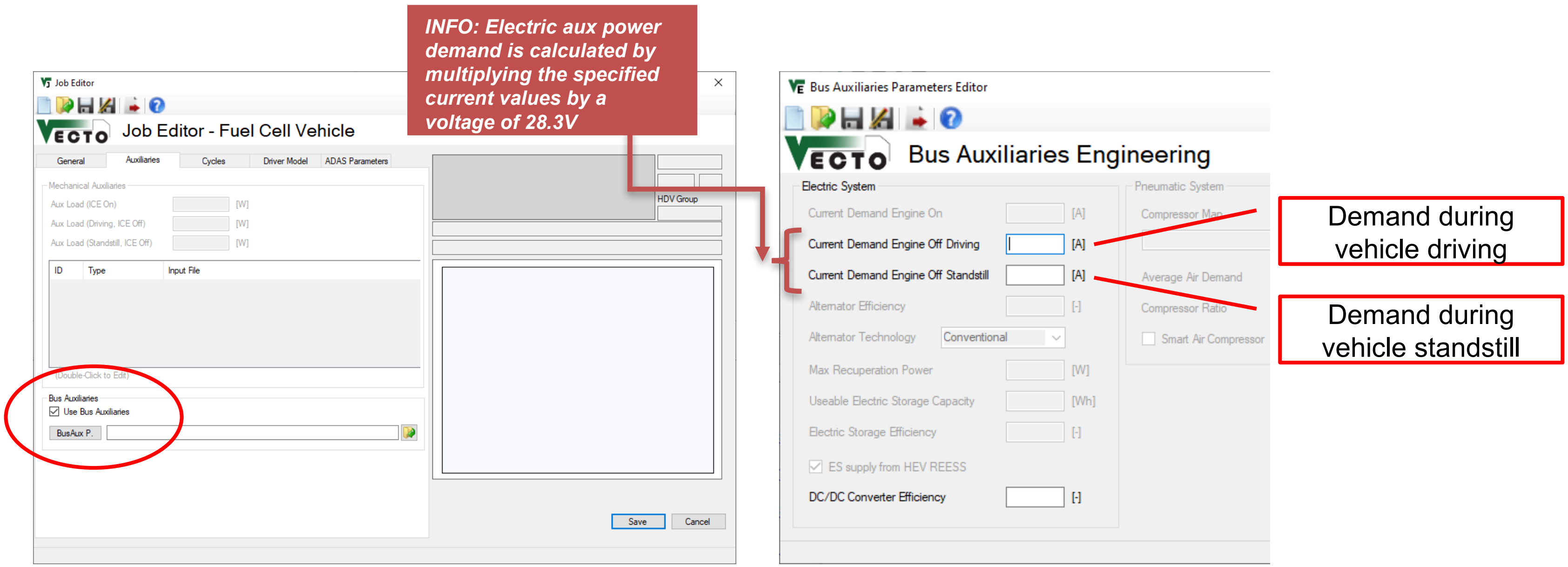
- CSV file format
  - 1st column: Electric power output of FCS in [kW]
  - 2nd column: H2 mass flow in [g/h]
- ***FCS component efficiency should also include DC/DC converter (as this is part of component test procedure)!***

```
P_el_out [kW],  m_H2 [g/h]
10.0,           1200.0
30.0,           2002.0
60.0,           3403.0
105.0,          5436.5
150.0,          8662.5
195.0,          12199.7
255.0,          17017.0
300.0,          21450.0
```

Example data

Fuel cell hybrid vehicles (FCHV) – VECTO features (5/5)

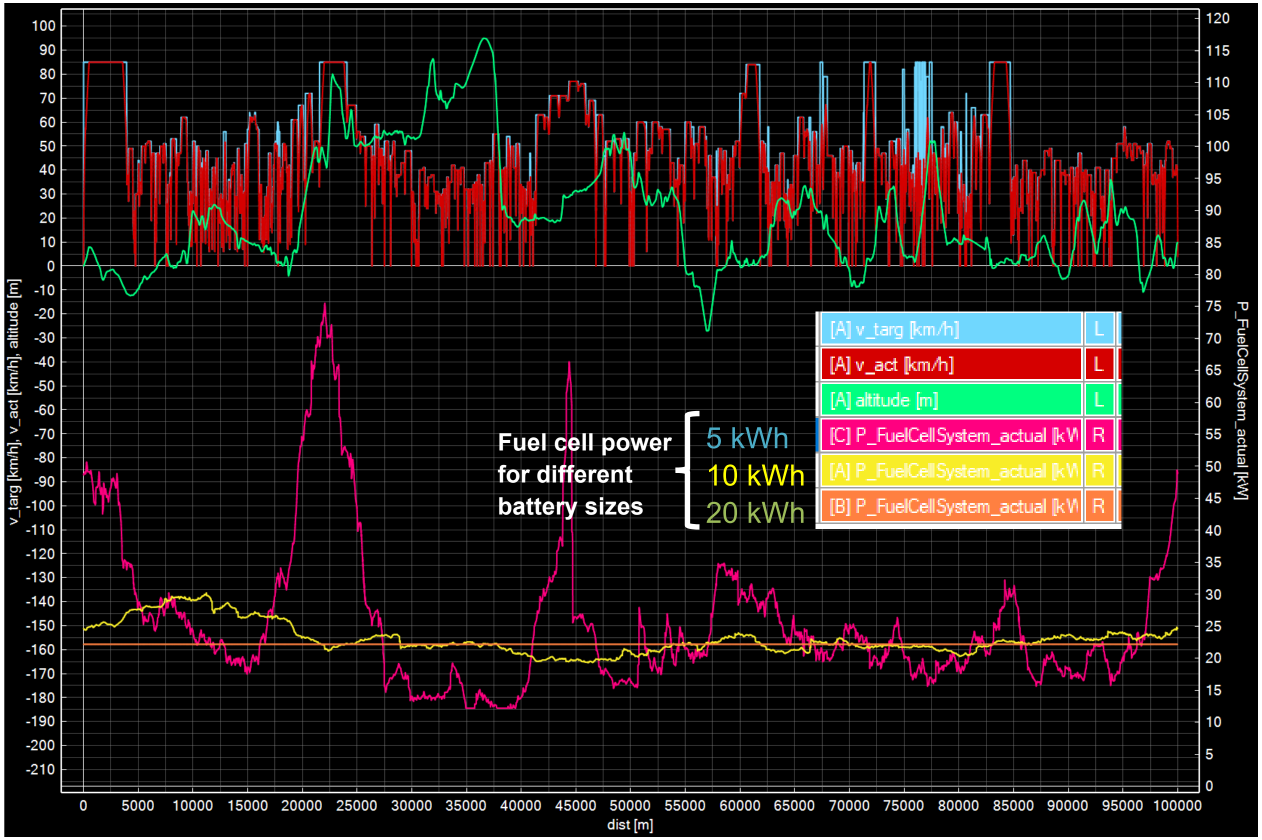
- Parameterization of auxiliaries for FCHV
  - Electric power demand of auxiliaries for both, lorries and buses, need to be parameterized via the “Bus Auxiliaries” GUI window



FCHV – Examples for generic operation strategy (1/5)

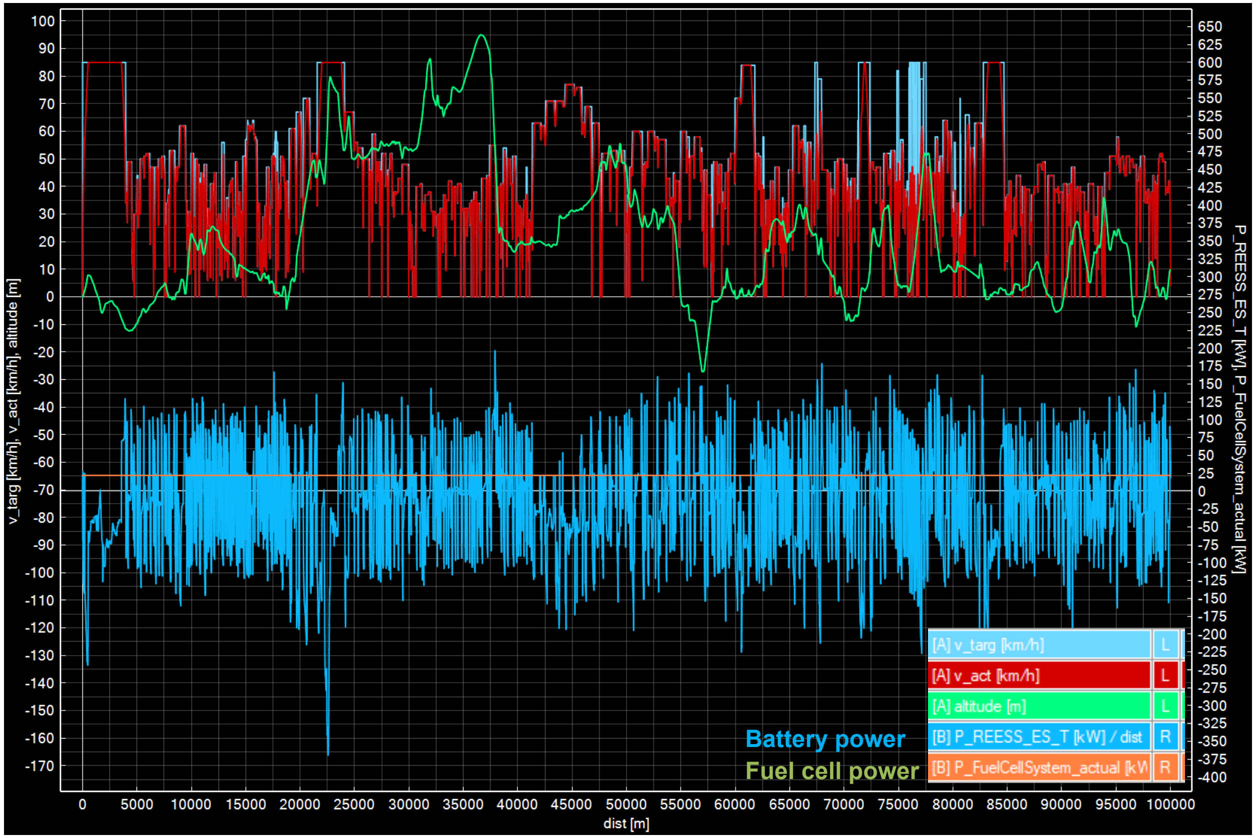
- Exemplary results for VECTO Group 2 12-ton rigid lorry (9.5 ton actual mass) on following slides
- Simulation in VECTO Urban Delivery Cycle
  - Average electric power demand around 22 kW
- FCHV powertrain specifications:
  - 100 kW FCS max power
  - following variants of battery:
    - 5 kWh
    - 10 kWh
    - 20 kWh
- Resulting speed profile was exactly the same for all different configurations

FCHV – Examples for generic operation strategy (2/5)



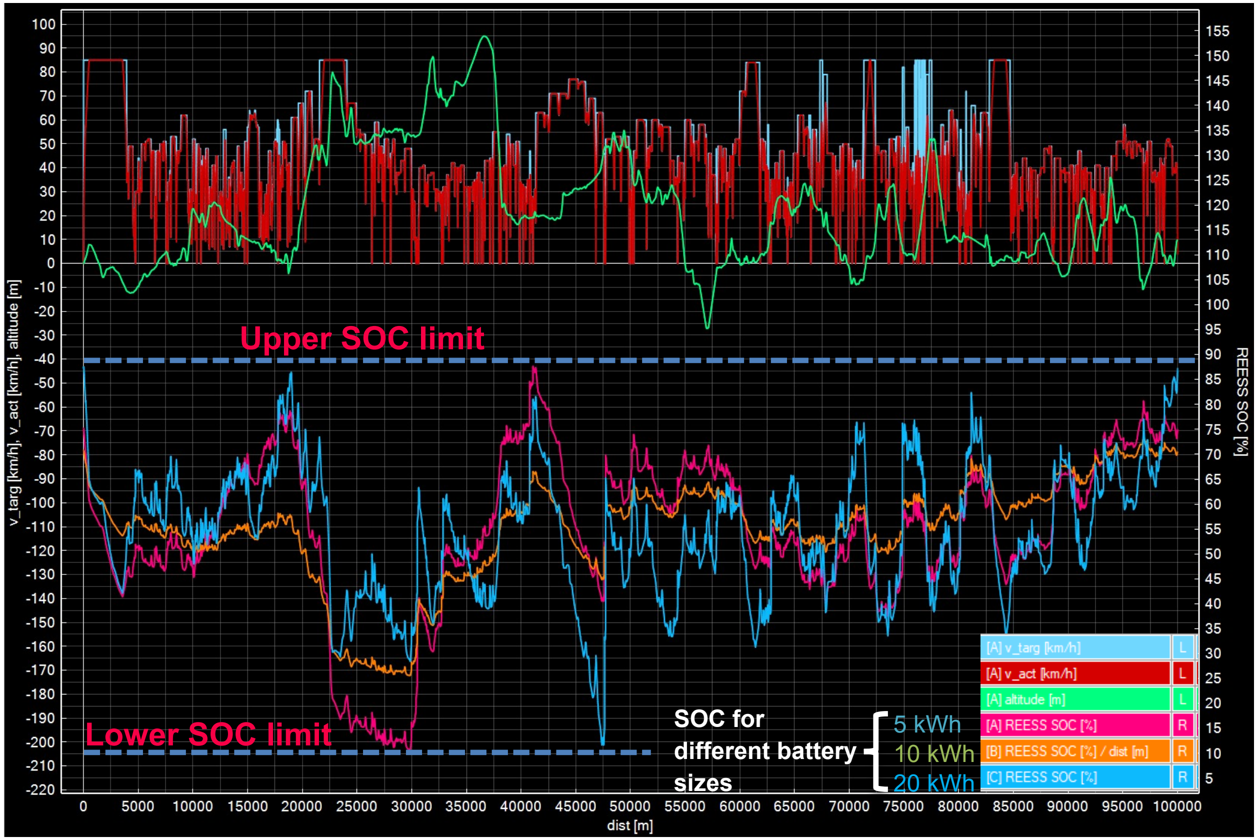
- **The lower the buffer size, the more dynamic the FCS operation gets**
- *Exactly same speed profile for all variants*
- Averaging window sizes determined:
  - 5 kWh: 6 km
  - 10 kWh: 37.5 km
  - 20 kWh: 100 km

FCHV – Examples for generic operation strategy (3/5)



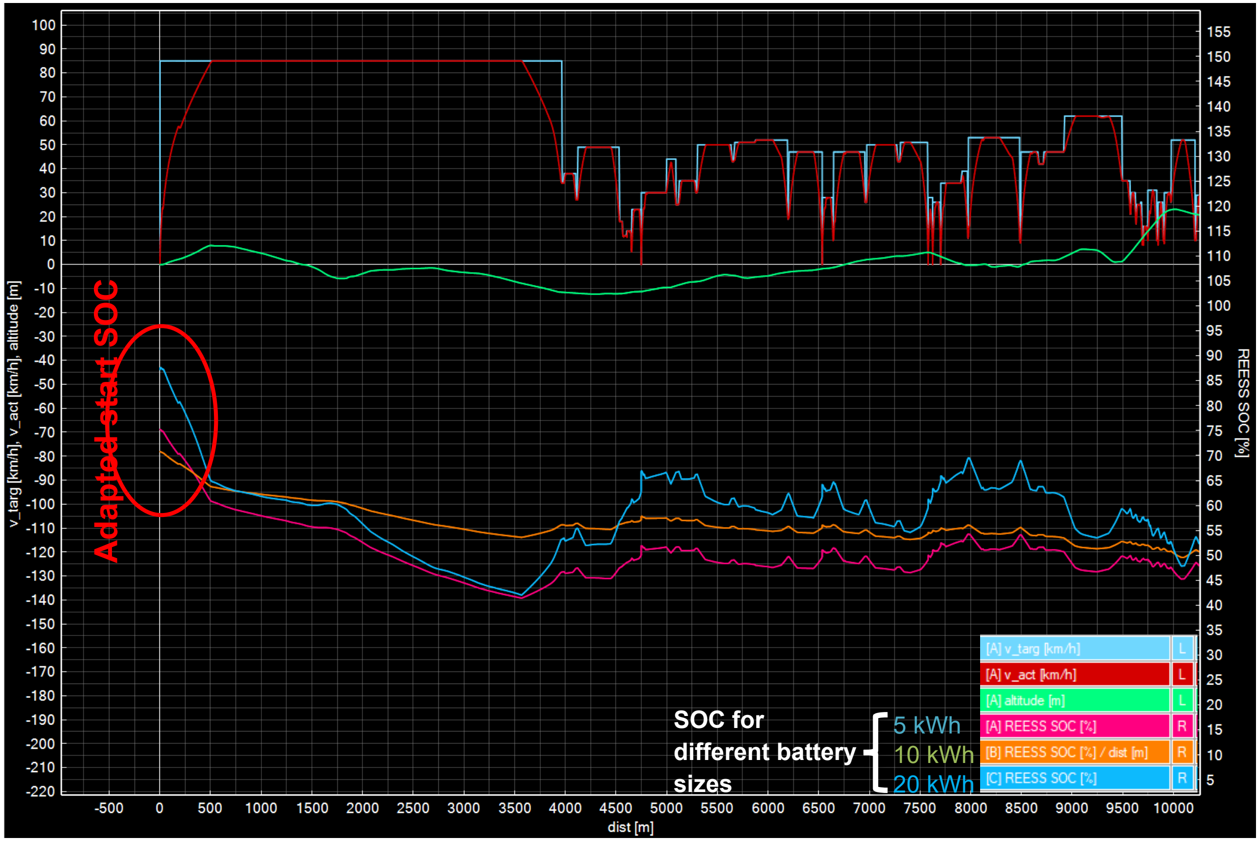
- Battery covers instantaneous deviations from average electric power demand
- Average electric power demand (i.e. FCS power) constant for the setting shown with 20 kWh REESS size

FCHV – Examples for generic operation strategy (4/5)



- **Using largest possible window size without violating SOC limits in any window → maximizing of battery buffer usage**
- *Exactly same speed profile for all variants*
- *SOC limits are Engineering settings, no connex to generic limits used in Declaration Mode*

FCHV – Examples for generic operation strategy (5/5)



- **Adaption of start SOC to specific mission in order to maximize usage of battery as buffer (detail from previous graph)**
- *Exactly same speed profile for all variants*

Fuel cell hybrid vehicles (FCHV) - Comments

- Release to be distributed via VECTO-DEV mailing list end of September
  - Basic description of FCHV features will also be added to VECTO release notes
  - Some example vehicles will be provided
- Current release restricted to one single FCS component
- **Updated release end of October will be able to handle:**
  - multiple FCS configurations as presented in TF FCS #17
    - Maximum 2x3 configuration
      i.e. 2 different FCS with a maximum number of 3 identical systems for each of the 2 different ones
  - Special provisions for handling power demands below minimum FCS power
  - IMC feature combined with FCHV

First VECTO beta release for the 3rd Amendment

Vehicles with In-Motion Charging Features

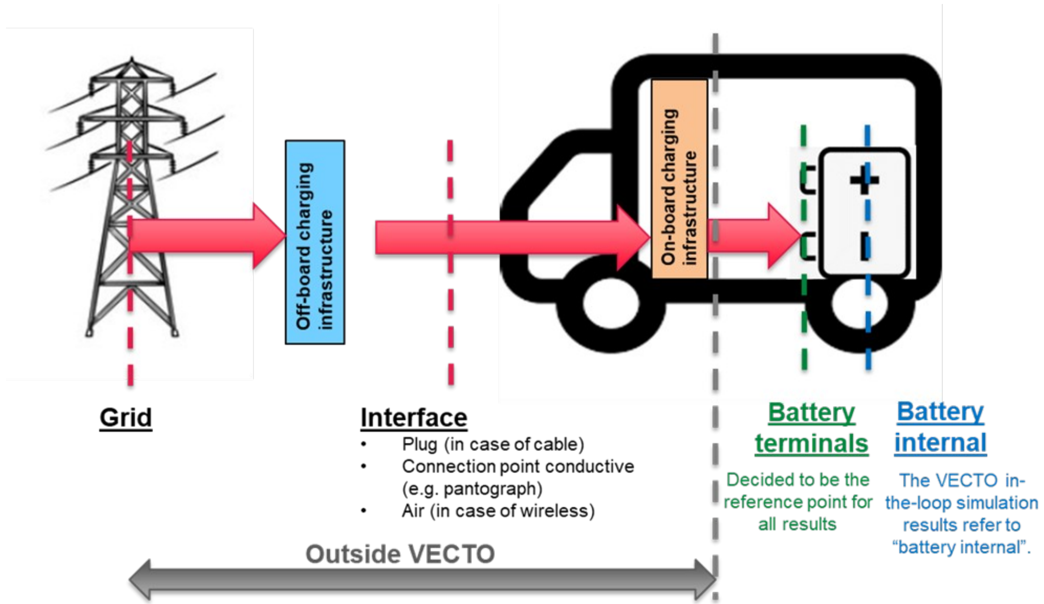
TUG

Vehicles with In-Motion Charging (IMC) features (1/11)

- One of the tasks of the pre-development project for the 3rd Amendment was to analyse different systems for in-use charging of electric vehicles
  (1) State of the art
  (2) Relevance and possible paths of implementation to Regulation (EU) 2017/2400
  (3) Functional prototype
- The analyses and proposals for (1) and (2) were presented and discussed with stakeholders in five stakeholder meetings (Oct 2021, Dec 2021, Jan 2022, Jun 2022 and Sept 2022).
- The following slides summarise the most important principles and present the functional prototype

Vehicles with In-Motion Charging (IMC) features (2/11)

- Reference point for results regarding electric energy consumption



- Principle applied already for the 2nd amendment
- Consequences for the 3rd amendment
  - Stationary charging: minor improvements to the Annex (max. charging power)
  - In-motion charging: Requires model extension and separate provisions in the Annex

Vehicles with In-Motion Charging (IMC) features (3/11)

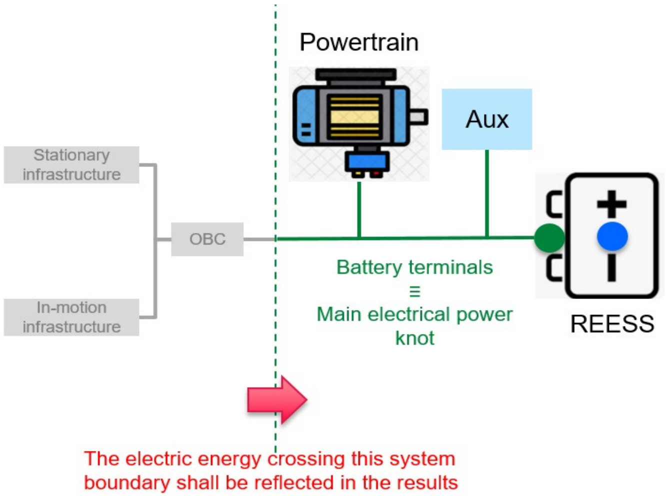
- Modelling in VECTO - Overview
  - ***Driving with "direct feed"*** incorporated to simulations of "charge depleting" mode by postprocessing (no additional sim time, no additional definitions etc.) (DECL)
    → higher utility factor / share of electric driving for OVC-HEV
    → (slightly) lower battery internal losses
  - ***Charging of batteries by IMC*** handled by postprocessing (DECL)
    → higher utility factor / share of electric driving for OVC-HEV
  - ***Impact on air drag*** incorporated to in-the loop simulation via a mission average share "active" (which applies either to motorway sections or to the total cycle) (ENG, DECL)
  - ***No direct influence on the "electric ranges"***, i.e. these are also only calculated on the basis of the usable battery capacity (DECL)

DECL / ENG: Effect visible in Declaration Mode / Engineering Mode

Report with detailed documentation to be published soon

Vehicles with In-Motion Charging (IMC) features (4/11)

- Modelling in VECTO – Calculations of electric energy consumption at battery terminals



Lila terms: new elements introduced for IMC modelling

$$EC_{el,terminal} = EC_{el,SOC} \cdot \left(S_{bat} \cdot \frac{1}{\eta_{bat(1)}} + S_{dir} \cdot \eta_{bat(2)} \right)$$

Where:

S_{bat} ..... Share of electric energy from grid first charged into battery

S_{dir} ..... Share of electric energy from grid used for direct feed

$\eta_{bat(1)}$ ... Eta_battery, average charging efficiency when charging from external sources

$\eta_{bat(2)}$ ... Eta_battery, average discharging efficiency during driving

Calculation examples see Task3_Masterexcel_OVC_IMC.xlsx as distributed with ppt
Report with detailed documentation to be published soon

Vehicles with In-Motion Charging (IMC) features (5/11)

- Modelling in VECTO – Current proposals for Declaration Mode

In-motion charging "technology"	Relevance			Proposal for definition in Regulation (EU) 2017/2400
	Heavy lorries	Heavy buses	Medium lorries	
Overhead pantograph	Yes	Yes	No	Vehicle equipped with overhead pantograph for connection with overhead catenary infrastructure as regulated by *standard tbd* and which is not overhead trolley.
Overhead trolley	No	Yes	No	Vehicle equipped with poles for connection with overhead catenary infrastructure as regulated in Annex 12 to UN Regulation 107 revision 8.
Ground-rail	Yes[1]	Yes[1]	Yes[1]	Charging technology that conductively transfers the electrical energy to the vehicle through rails embedded in or on top of the road surface.
Wireless	Yes[1]	Yes[1]	Yes[1]	Charging technology that inductively transfers the electrical energy to the vehicle through devices embedded in or on top of the road surface providing magnetic fields as they are specified IEC 61980.

[1] Only Potential niche applications are conceivable.

Eta: to be worked out for the 3rd amendment

Vehicles with In-Motion Charging (IMC) features (6/11)

- Modelling in VECTO – Proposal for IMC infrastructure availability

Overhead - Pantograph

Vehicle group	Share of distance of total mission where in-motion charging infrastructure is available - Long haul	Regional delivery	Urban delivery	Muni-cipal	Con-struction	Heavy Urban	Urban	Sub-urban	Inter-urban	Coach
53	---	0.00	0.00	---	---	---	---	---	---	---
54	---	0.00	0.00	---	---	---	---	---	---	---
1s	---	0.34	0.00	---	---	---	---	---	---	---
1	---	0.34	0.00	---	---	---	---	---	---	---
2	0.50	0.34	0.00	---	---	---	---	---	---	---
3	---	0.34	0.00	---	---	---	---	---	---	---
4	0.50	0.34	0.00	0.00	0.00	---	---	---	---	---
5	0.50	0.34	0.00	---	0.00	---	---	---	---	---
9	0.50	0.34	---	0.00	0.00	---	---	---	---	---
10	0.50	0.34	---	---	0.00	---	---	---	---	---
11	0.50	0.34	---	0.00	0.00	---	---	---	---	---
12	0.50	0.34	---	---	0.00	---	---	---	---	---
16	---	---	---	---	0.00	---	---	---	---	---
31	---	---	---	---	---	0.30	0.30	0.30	0.30	---
32	---	---	---	---	---	---	---	---	0.30	0.30
33	---	---	---	---	---	0.30	0.30	0.30	0.30	---
34	---	---	---	---	---	---	---	---	0.30	0.30
35	---	---	---	---	---	0.30	0.30	0.30	0.30	---
36	---	---	---	---	---	---	---	---	0.30	0.30
37	---	---	---	---	---	0.30	0.30	0.30	0.30	---
38	---	---	---	---	---	---	---	---	0.30	0.30
39	---	---	---	---	---	0.30	0.30	0.30	0.30	---
40	---	---	---	---	---	---	---	---	0.30	0.30

Overhead - Trolley

Vehicle group	Share of distance of total mission where in-motion charging infrastructure is available - Long haul	Regional delivery	Urban delivery	Muni-cipal	Con-struction	Heavy Urban	Urban	Sub-urban	Inter-urban	Coach
53	---	0.00	0.00	---	---	---	---	---	---	---
54	---	0.00	0.00	---	---	---	---	---	---	---
1s	---	0.00	0.00	---	---	---	---	---	---	---
1	---	0.00	0.00	---	---	---	---	---	---	---
2	0.00	0.00	0.00	---	---	---	---	---	---	---
3	---	0.00	0.00	---	---	---	---	---	---	---
4	0.00	0.00	0.00	0.00	0.00	---	---	---	---	---
5	0.00	0.00	0.00	---	0.00	---	---	---	---	---
9	0.00	0.00	---	0.00	0.00	---	---	---	---	---
10	0.00	0.00	---	---	0.00	---	---	---	---	---
11	0.00	0.00	---	0.00	0.00	---	---	---	---	---
12	0.00	0.00	---	---	0.00	---	---	---	---	---
16	---	---	---	---	0.00	---	---	---	---	---
31	---	---	---	---	---	0.80	0.80	0.80	0.20	---
32	---	---	---	---	---	---	---	---	0.20	0.00
33	---	---	---	---	---	0.80	0.80	0.80	0.20	---
34	---	---	---	---	---	---	---	---	0.20	0.00
35	---	---	---	---	---	0.80	0.80	0.80	0.20	---
36	---	---	---	---	---	---	---	---	0.20	0.00
37	---	---	---	---	---	0.80	0.80	0.80	0.20	---
38	---	---	---	---	---	---	---	---	0.20	0.00
39	---	---	---	---	---	0.80	0.80	0.80	0.20	---
40	---	---	---	---	---	---	---	---	0.20	0.00

Vehicles with In-Motion Charging (IMC) features (7/11)

- Modelling in VECTO – Proposal for IMC infrastructure availability

Ground rail + Wireless

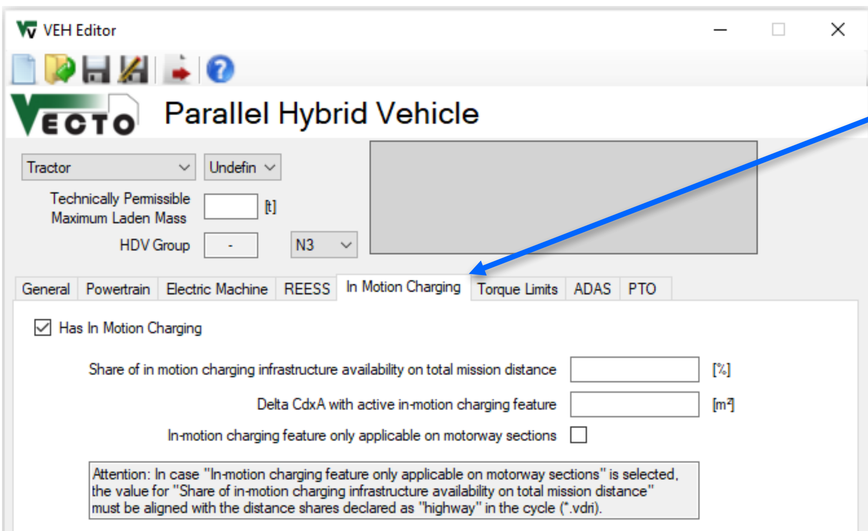
Vehicle group	Share of distance of total mission where in-motion charging infrastructure is available -									
	Long haul	Regional delivery	Urban delivery	Muni-cipal	Con-struction	Heavy Urban	Urban	Sub-urban	Inter-urban	Coach
53	---	0.00	0.50	---	---	---	---	---	---	---
54	---	0.00	0.50	---	---	---	---	---	---	---
1s	---	0.00	0.50	---	---	---	---	---	---	---
1	---	0.00	0.50	---	---	---	---	---	---	---
2	0.00	0.00	0.50	---	---	---	---	---	---	---
3	---	0.00	0.50	---	---	---	---	---	---	---
4	0.00	0.00	0.50	0.50	0.00	---	---	---	---	---
5	0.00	0.00	0.50	---	0.00	---	---	---	---	---
9	0.00	0.00	---	0.50	0.00	---	---	---	---	---
10	0.00	0.00	---	---	0.00	---	---	---	---	---
11	0.00	0.00	---	0.50	0.00	---	---	---	---	---
12	0.00	0.00	---	---	0.00	---	---	---	---	---
16	---	---	---	---	0.00	---	---	---	---	---
31	---	---	---	---	---	0.50	0.50	0.50	0.00	---
32	---	---	---	---	---	---	---	---	0.00	0.00
33	---	---	---	---	---	0.50	0.50	0.50	0.00	---
34	---	---	---	---	---	---	---	---	0.00	0.00
35	---	---	---	---	---	0.50	0.50	0.50	0.00	---
36	---	---	---	---	---	---	---	---	0.00	0.00
37	---	---	---	---	---	0.50	0.50	0.50	0.00	---
38	---	---	---	---	---	---	---	---	0.00	0.00
39	---	---	---	---	---	0.50	0.50	0.50	0.00	---
40	---	---	---	---	---	---	---	---	0.00	0.00

Vehicles with In-Motion Charging (IMC) features (8/11)

- Modelling in VECTO – Further proposed assumptions
  - Battery charging during mission
    - 50% of usable battery capacity via “SOC lift” by IMC
    - No further stationary charging
  - Air drag
    - Considered only for “Overhead – pantograph”
    - $\Delta CdxA = 0.6\ m^2$ for the shares in the mission where IMC is available

Vehicles with In-Motion Charging (IMC) features (9/11)

- Inputs in Graphical User Interface (ENG)



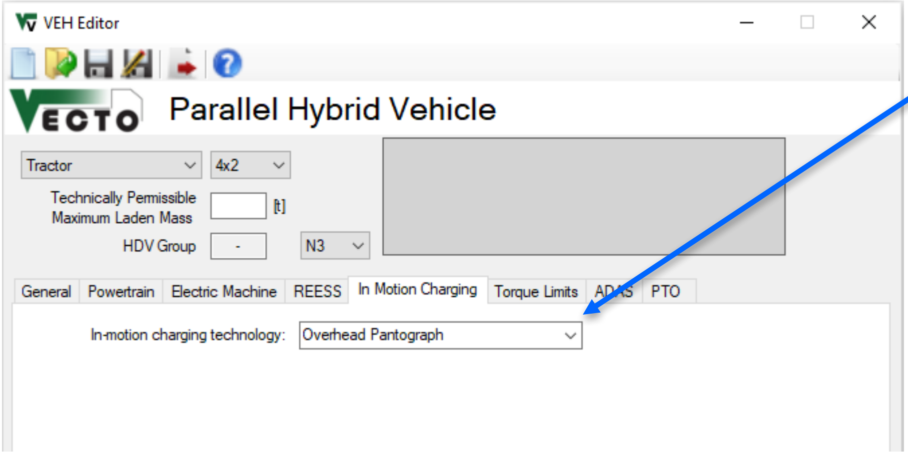
New tab in “VEH Editor” for HEV and PEV vehicles

Attention!

- Working with ENG mode simulations for IMC features requires detailed knowledge of the proposed approach (mainly for developers)
- For checking of regulatory impact DECL mode testing recommended!

Vehicles with In-Motion Charging (IMC) features (10/11)

- Inputs in Graphical User Interface (DECL)



Dropdown Menu with pre-defined technologies
- None
- Overhead Pantograph
- Overhead Trolley
- Ground Rail
- Wireless

Vehicles with In-Motion Charging (IMC) features (11/11)

- Example vehicle models will be distributed with the release (located in “\Generic vehicles”)
- Base specs example models
  - Group 5 / HEV P2 / OVC: P_{cont_EM}: 200kW | E_{Reess}: 120kWh nominal
  - Group 5 / PEV: P_{cont_EM}: 325kW | E_{Reess}: 170kWh nominal
- Example results: Long haul with rep payload
  Comparison results “w/o” and “w/” pantograph with similar other vehicle specs like mass
  - Group 5 / P2 / OVC w/o pantograph: UF = 0.149; $CO2_{weighted}$ = 692.2 g/km
  - Group 5 / P2 / OVC w/ pantograph: UF = 0.660; $CO2_{weighted}$ = 347.1 g/km → **-50% CO2**

  - Group 5 / PEV w/o pantograph: EC_{el} = 1.262 kWh/km
  - Group 5 / PEV w/ pantograph: EC_{el} = 1.297 kWh/km → +2.8% EC_{el}